

Generalized Probabilistic Topological Field Theories: Summary

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Conceptual skeleton of the GPTFT program: the spacetime axioms, topological field theories as finite algebraic data, the category of GPTs as **Vec** plus positivity, the 1d case as dualizability, and the 2d case as a commutative Frobenius algebra whose finite data are positive GPT data. Existence of a 2d theory reduces to an intrinsic condition on the state space — an *operational Frobenius structure*, whose physically central case is a *copyable maximal frame* of perfectly distinguishable states. *Spectral mountability* (every frame copyable, coherently) derives the dagger-compact postulates of the Selby–Scandolo–Coecke reconstruction, making spacetime-mountability a candidate reconstruction axiom.

THE SPACETIME AXIOMS

- A field theory assigns: $\text{space} \rightarrow \text{system}$, $\text{spacetime} \rightarrow \text{process}$.
- Locality is the compatibility of these assignments:

$$Z(M' \circ M) = Z(M') \circ Z(M), \quad Z(\Sigma \sqcup \Sigma') = Z(\Sigma) \otimes Z(\Sigma').$$

Gluing $\rightarrow$ composition (temporal locality); disjoint union $\rightarrow$ joint system (spatial locality).

- Axioms (Atiyah–Segal [1, 2]): a field theory is a symmetric monoidal functor

$$Z : \mathbf{Bord}_n \longrightarrow \mathcal{C}.$$

- The source is a choice of what spacetime is: in a *topological* theory, bordisms up to diffeomorphism — no metric, no local excitations.
- The target is a choice of what physics is: **Vec**, **Hilb** encode the linear skeleton of quantum theory; a *GPT* target encodes directly the operational, probabilistic content [3–6].
- **GPTFT** := TFT valued in the category of GPTs. Question: *which probabilistic theories admit one?*

TOPOLOGICAL FIELD THEORIES

- $\mathbf{Bord}_n$: objects = closed oriented $(n - 1)$ -manifolds; morphisms = compact oriented n -bordisms up to diffeomorphism rel boundary; composition = gluing; $(\sqcup, \emptyset)$ the monoidal structure.
- Diffeomorphism invariance $\Rightarrow$ every manifold decomposes into finitely many elementary pieces $\Rightarrow$ a TFT is *finite algebraic data* in $\mathcal{C}$: values on generators, subject to the relations identifying different decompositions of the same manifold.
- $d = 1$: generators interval, cup, cap; relations the snakes; a theory = a *dualizable object* of $\mathcal{C}$ (cobordism hypothesis [8]).
- $d = 2$: generators disk, pants, copants, reversed disk; a theory = a *commutative Frobenius algebra* in $\mathcal{C}$ [16–18].
- Closed manifolds $\rightarrow$ endomorphisms of **1**, i.e. scalars: the *invariants* of the theory.

THE CATEGORY OF GPTS: VEC PLUS POSITIVITY

Vec (finite-dimensional real): all linear maps, monoidal product $\otimes$, compact closed — duals, ev, coev canonical, snakes automatic. GPT is the same data with one positivity condition per layer:

$$\begin{aligned} V &\rightsquigarrow (V, V^+, u), \\ f &\rightsquigarrow f \text{ with } f(V_A^+) \subseteq V_B^+, \\ V_A \otimes V_B &\rightsquigarrow V_{AB}^+ \in [\otimes_{\min}, \otimes_{\max}], \quad u_A \otimes u_B. \end{aligned}$$

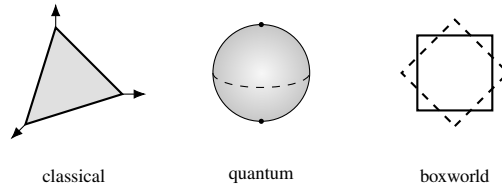


FIG. 1. Normalized-state slices. Classical: simplex (self-dual orthant). Quantum: Bloch ball (self-dual PSD cone). Boxworld: square, dual slice dashed — not self-dual.

- *Objects*: systems $A = (V_A, V_A^+, u_A)$; V_A^+ a proper cone of unnormalized states, $u_A \in V_A^*$ strictly positive on it. Normalized states $\Omega_A = u_A^{-1}(1) \cap V_A^+$; effects in the dual cone $(V_A^+)^*$; allowed effects $[0, u_A]$; measurements (e_i) , $\sum_i e_i = u_A$. No-restriction by default.
- *Morphisms*: cone-preserving linear maps. $\text{Hom}_{\text{GPT}}(A, B)$ is a *cone*, not a subspace; GPT is a wide, non-full subcategory of **Vec**. States and effects are arrows $\mathbf{1} \rightarrow A$, $A \rightarrow \mathbf{1}$; probabilities are composites $e \circ f \circ \omega \in \text{End}(\mathbf{1}) = \mathbb{R}$.
- *Composites*: underlying space $V_A \otimes V_B$ (tomographic locality), unit $u_A \otimes u_B$, composite cone any proper cone in $[\otimes_{\min}, \otimes_{\max}]$. The interval collapses iff one factor is classical [7]; the quantum cone sits strictly between.
- *Departure from Vec*: duals survive on objects ($X^\vee$ swaps state and effect cones onto V^*), but *ev* and *coev* are morphisms only if *positive*. Compact closure — automatic in **Vec** — becomes a contingent property of the cones.

Examples (Fig. 1): classical $(\mathbb{R}^N, \mathbb{R}_{\geq 0}^N, \sum_i)$, quantum $(H_N(\mathbb{C}), \text{PSD}, \text{tr})$, boxworld (cone over the square [4]). Self-duality of the state cone is the property that will decide everything below.

ONE DIMENSION: DUALIZABILITY

- **Bord**₁: generated by id_+ , $\text{cup } \cup : \emptyset \rightarrow + \sqcup -$, $\text{cap } \cap : - \sqcup + \rightarrow \emptyset$, modulo the snakes; $S^1 = \cap \circ \sigma \circ \cup$. A 1d theory is a dualizable object $X = Z(+)$ [8].
- In **Vec**: every object dualizable, $\text{coev}(1) = g := \sum_a f_a \otimes f^a$, snakes = resolution of identity, $Z(S^1) = \dim_{\mathbb{R}} V$. Classification: one integer. Nothing to check.
- In GPT: the dual is *forced*, $X^\vee = (V^*, (V^+)^*, u^\vee)$, and dualizability of $X = (V, V^+, u)$ reduces to two memberships of the *single* element g :

$$\text{cup (state): } g \in V_{X \otimes X^\vee}^+, \quad (1)$$

$$\text{cap (effect): } g \in (V_{X^\vee \otimes X}^+)^*. \quad (2)$$

The cup wants the composite cone large, the cap wants it small; min–max duality exchanges the two ends. Operationally, (1)–(2) assert exact teleportation with g as both resource and measurement [9].

- *Conjecture*: (1)–(2) are satisfiable iff V^+ is *weakly self-dual*. Classical: passes, g separable ($\in \otimes_{\min}$). Quantum: passes, g = maximally entangled element after a transpose twist. Boxworld: no-go (square cone not self-dual; only finitely many twists).
- Koecher–Vinberg [10] + BBLW steering [9]: homogeneous self-dual cones are Euclidean Jordan algebras $\Rightarrow$ theories passing the 1d test are “close to quantum.”

TWO DIMENSIONS: THE FROBENIUS DATA AS GPT DATA

A 2d theory assigns $X := Z(S^1) = (V, V^+, u)$ and one morphism per generator (Fig. 2); the relations (unit, counit, (co)associativity, (co)commutativity, Frobenius) are equalities of linear maps, unchanged from **Vec**. The finite data, read in GPT:

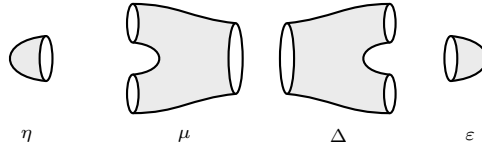


FIG. 2. Generators of $\mathbf{Bord}_2$: disk η , pants μ , copants Δ , reversed disk ε (time flows left to right).

bordism	algebra	GPT positivity
disk $\eta : \emptyset \rightarrow S^1$	unit 1_A	a state: $\eta(1) \in V^+$
rev. disk $\varepsilon : S^1 \rightarrow \emptyset$	counit	an effect: $\varepsilon \in (V^+)^*$
pants $\mu : S^1 \sqcup S^1 \rightarrow S^1$	product	$\mu(V_{XX}^+) \subseteq V^+$
copants $\Delta : S^1 \rightarrow S^1 \sqcup S^1$	coproduct	$\Delta(V^+) \subseteq V_{XX}^+$

Definition (2d GPTFT, finite data). A GPT system $X = (V, V^+, u)$, a composite cone $V_{XX}^+ \in [\otimes_{\min}, \otimes_{\max}]$, and four positive structure maps $\eta, \varepsilon, \mu, \Delta$ as in the table, satisfying the (co)unit, (co)associativity, (co)commutativity and Frobenius relations. Equivalently: a commutative Frobenius algebra in $\mathbf{Vec}$ all of whose structure maps are positive for the chosen cones.

Conceptual points.

- *Operational reading.* The unit is a preparation (a distinguished unnormalized state created by the disk); the counit is a measurement outcome; merging (μ) and splitting (Δ) of circles are physical transformations.
- *The cone tension returns.* μ is easiest to make positive when V_{XX}^+ is *small* (fewer inputs to control), Δ when it is *large* (more room for outputs): the 2d incarnation of the 1d cup/cap tension, now on genuinely independent data.
- *The 1d problem embeds.* The derived pairing and copairing

$$\beta := \varepsilon \circ \mu \in (V_{XX}^+)^*, \quad \gamma := \Delta(\eta(1)) \in V_{XX}^+$$

are a positive effect and a positive state, and the Frobenius + (co)unit relations force the snake identities for (γ, β) . Hence X is dualizable with $X^\vee \cong X$: the 2d structure contains a solution of the 1d problem *with the dual identified with the object itself*.

- *Self-duality becomes necessary.* Since the dual is forced to be $X^\vee = (V^*, (V^+)^*, u^\vee)$, the isomorphism $X \cong X^\vee$ is a positive linear isomorphism $V \cong V^*$ carrying V^+ onto $(V^+)^*$: *weak self-duality of the state cone is a necessary condition for a 2d GPTFT*. What is conjectural in 1d is forced in 2d. Immediate corollary: boxworld admits no 2d GPTFT.
- *Invariants are nonnegative.* The handle operator $h := \mu \circ \Delta$ is a positive endomorphism of X , and

$$Z(\Sigma_g) = \varepsilon(h^g(\eta(1))) \geq 0 :$$

every closed surface evaluates to a composition of positive data — an (unnormalized) probability. In particular $Z(S^2) = \varepsilon(\eta(1)) \geq 0$ and $Z(T^2) = \dim_{\mathbb{R}} V$.

- *Classical passes — twice over.* Basis type: the pointwise algebra on $\mathbb{R}^N$, $\eta(1) = \sum_i e_i$, $\varepsilon(e_i) = \lambda_i > 0$, $\mu(e_i \otimes e_j) = \delta_{ij} e_i$, $\Delta(e_i) = \lambda_i^{-1} e_i \otimes e_i$; one family per weight vector λ , with invariants $Z(\Sigma_g) = \sum_i \lambda_i^{1-g}$. Gauge type (no copyable frame): the nilpotent algebra $\mathbb{R}[x]/(x^2)$ on the bit cone $\text{cone}\{1, x\}$ with $\varepsilon = s x^*$ ($Z(S^2) = 0$, $Z(T^2) = 2$), and group algebras $\mathbb{R}[G]$ with cone over the group elements — 2d lattice gauge theories mounted on classical GPTs, whose would-be frames (Fourier idempotents) are not physical states.
- *Quantum passes — the coherent copy.* For each orthonormal basis, with $W = \sum_i |ii\rangle\langle i|$:

$$\Delta(\rho) = W \rho W^\dagger, \quad \mu(x) = W^\dagger x W \quad (\text{Schur product}), \quad \eta(1) = J = \sum_{ij} |i\rangle\langle j|, \quad \varepsilon = \langle \eta | \cdot | \eta \rangle.$$

All four maps are positive for the standard quantum composite (positivity of μ is the Schur product theorem); the derived cup is the maximally entangled $\gamma = |\Phi\rangle\langle\Phi|$ — realizing the transpose twist conjectured in the 1d analysis — the handle is trivial ($h = \text{id}$, the structure is special), and $Z(\Sigma_g) = N^2$ for every genus. No-cloning is respected: the *trace*-marginals of Δ give dephasing; only the ε -marginal is lossless.

- *No-broadcasting dichotomy.* If ε is strictly positive, the counit laws make Δ a universal broadcaster for the system (V, V^+, ε) , and the generalized no-broadcasting theorem [12] forces a simplicial cone. Nonclassical solutions therefore have *boundary* counit — in quantum, the pure effect $\langle \eta | \cdot | \eta \rangle$. Diagrammatically [11]: purity of the cup $\iff$ triviality of leaks $\iff$ fully quantum.

THE EXISTENCE CONDITION: COPYABLE FRAMES

Reduction to the state space. All the data of a 2d GPTFT collapse onto a product and one effect:

Proposition 1 (Operational Frobenius structure). $X = (V, V^+, u)$ admits a 2d GPTFT iff V carries a commutative associative product $\star$ with unit η , and an effect $\varepsilon \in (V^+)^*$, such that: (i) $\eta \in V^+$ and $V^+ \star V^+ \subseteq V^+$ — the cone is closed under the product; (ii) $\beta(v, w) := \varepsilon(v \star w)$ is nondegenerate and $\flat : v \mapsto \beta(v, -)$ is an order isomorphism $(V, V^+) \cong (V^*, (V^+)^*)$ — the effect self-dualizes the cone through the product; (iii) the handle element $H = \sum_a b_a \star b^a$ (any basis, β -dual basis) acts positively. The coproduct and counit are then derived, $\Delta = \mu^\dagger$, $\varepsilon = \eta^\dagger$, and the composite cone can be taken minimal: $V_{XX}^+ = \overline{\text{cone}}(\otimes_{\min} \cup \Delta(V^+))$.

Dagger form (under tomographic locality). β induces a dagger $f^\dagger := \flat^{-1} \circ f^* \circ \flat$ on the monoidal subcategory generated by X ; condition (ii) says precisely that this dagger *preserves positivity*. A 2d GPTFT is then a positive dagger together with a commutative $\dagger$ -Frobenius algebra (μ, η) with μ, η positive — the GPT form of the correspondence between $\dagger$ -Frobenius algebras, orthonormal bases, and finite-dimensional C^* -algebras [13, 14]. Two bilinear forms coexist and divide the labor:

$$\beta = \varepsilon \circ \mu, \quad \beta_s(v, w) = \varepsilon(v \star Tw),$$

where T is the canonical *Galois involution* of the algebra (identity on the real spectrum, conjugation on the complex spectrum; in quantum, $\rho \mapsto \rho^T$). The copairing of β is the positive cup — the teleportation resource; in quantum $\gamma = |\Phi\rangle\langle\Phi|$, $\beta = \text{tr}(\rho \sigma^T)$. The copairing of β_s is *not* positive (in quantum, SWAP), but β_s is a GNS-type inner product whose adjoint is the *sharp dagger* (in quantum, Hilbert–Schmidt, giving the Hermitian adjoint). Cups give the transpose; the Frobenius algebra supplies the conjugation; their composite is the dagger — $(\star, T, \varepsilon)$ is an abstract commutative $*$ -algebra with a positive functional, compatible with the cone.

The operational condition.

Definition (Copyable frame). A maximal set of perfectly distinguishable states $\{\omega_i\}_{i=1}^n$ of X is copyable if there exist an admissible composite cone, a positive linear map $\Delta : X \rightarrow X \otimes X$, and an effect $\varepsilon \in (V^+)^*$ with: (C1) copying: $\Delta(\omega_i) = \omega_i \otimes \omega_i$; (C2) losslessness: $\varepsilon(\omega_i) = 1$ and $(\varepsilon \otimes \text{id})\Delta = \text{id} = (\text{id} \otimes \varepsilon)\Delta$; (C3) consistency: Δ is coassociative and cocommutative; (C4) mergeability: the copairing dual to the induced β equals $\Delta(\eta)$ for a state $\eta \in V^+$, and the induced merge μ is positive.

Definition (Spectrally mountable). X is spectrally mountable if every maximal set of perfectly distinguishable states is copyable, and the daggers induced by the twisted forms β_s of the resulting structures coincide (coherence).

Proposition 2 (Existence). A single copyable frame yields a 2d GPTFT of basis type: the frame is the positive real spectrum of the algebra, and the twisted dagger is sharp at it. Every basis-type theory arises this way. The inclusion in all 2d GPTFTs is strict: gauge-type solutions exist with no copyable frame (nilpotent and group-algebra structures above). Every 2d GPTFT has at least one real character (Perron–Frobenius applied to the commuting positive family $L_v = v \star (-)$).

Proposition 3 (Alignment with the SSC reconstruction [11]). If X is spectrally mountable in a tomographically local GPT, then Postulates 1–4 of [11] hold for the process theory generated by X : (1) process theory and (2) finite local classical interface are the ambient framework; (3) cups & caps are $\gamma = \Delta(\eta)$ and $\beta = \varepsilon \circ \mu$; (4) the common twisted dagger is sharp at every maximal testable preparation, since every frame is copyable. By the lemmas of [11], X is then spectral. Their Postulate 5 (essentially unique symmetric purification $\Rightarrow$ homogeneity $\Rightarrow$ Koecher–Vinberg $\Rightarrow$ Euclidean Jordan algebras) is the remaining distance to quantum $\oplus$ classical theory.

Quantum theory is spectrally mountable: one coherent copier per orthonormal basis, and all twisted forms collapse to the basis-independent Hilbert–Schmidt inner product, whose adjoint is the Hermitian dagger. Classical theory is spectrally mountable through the pointwise copiers. Boxworld fails already at the derived 1d layer.

The ladder. Each condition strictly refines the last (as classes of GPTs):

$$\{\text{teleportable}\} \supseteq \{\text{mountable}\} \supseteq \{\text{copyable frame}\} \supseteq \{\text{spectrally mountable}\} \supseteq \{\text{quantum} \oplus \text{classical}\},$$

where *teleportable* = admits a 1d GPTFT (cups & caps; weak self-duality), *mountable* = admits some 2d GPTFT (operational Frobenius structure; at least one real character; nonnegative invariants), *copyable frame* = admits a basis-type theory, and *spectrally mountable* as above ($\Rightarrow$ SSC Postulates 1–4 $\Rightarrow$ spectrality). The last inclusion is conjectured to be an equality up to the Jordan-algebraic loose ends.

Open problems. (1) *Coherence*: is the common-dagger condition automatic when every frame is copyable? (2) *Converse of Prop. 3*: can the coherent copier be constructed from a sharp dagger — the “spiders” question raised in the outlook of [11]? (3) *Spin factors*: are the d -ball systems ($d \neq 3$) mountable? Decisive for whether 2d mountability alone cuts the Jordan family to $\mathbb{C}$ -quantum $\oplus$ classical. (4) The TFT origin of symmetric purification (SSC Postulate 5). (5) The non-tomographically-local ($C \neq 0$, fermionic) case.

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- [1] M. F. Atiyah, *Topological quantum field theories*, Publ. Math. IHÉS **68**, 175 (1988).
- [2] G. Segal, *The definition of conformal field theory*, in *Topology, Geometry and Quantum Field Theory*, London Math. Soc. Lecture Note Ser. **308** (Cambridge University Press, 2004).
- [3] L. Hardy, *Quantum theory from five reasonable axioms*, arXiv:quant-ph/0101012.
- [4] J. Barrett, *Information processing in generalized probabilistic theories*, Phys. Rev. A **75**, 032304 (2007), arXiv:quant-ph/0508211.
- [5] M. P. Müller, *Probabilistic theories and reconstructions of quantum theory*, SciPost Phys. Lect. Notes **28** (2021), arXiv:2011.01286.
- [6] M. Plávala, *General probabilistic theories: An introduction*, Phys. Rep. **1033**, 1 (2023), arXiv:2103.07469.
- [7] G. Aubrun, L. Lami, C. Palazuelos, and M. Plávala, *Entangleability of cones*, Geom. Funct. Anal. **31**, 181 (2021), arXiv:1911.09663.
- [8] J. Lurie, *On the classification of topological field theories*, Current Developments in Mathematics **2008**, 129 (Int. Press, 2009), arXiv:0905.0465.
- [9] H. Barnum, J. Barrett, M. Leifer, and A. Wilce, *Teleportation in general probabilistic theories*, Proc. Sympos. Appl. Math. (AMS, 2012), arXiv:0805.3553.
- [10] M. Koecher, *The Minnesota Notes on Jordan Algebras and Their Applications*, Lecture Notes in Mathematics **1710** (Springer, 1999); E. B. Vinberg, *The theory of convex homogeneous cones*, Trans. Moscow Math. Soc. **12**, 340 (1963).
- [11] J. H. Selby, C. M. Scandolo, and B. Coecke, *Reconstructing quantum theory from diagrammatic postulates*, Quantum **5**, 445 (2021), arXiv:1802.00367.
- [12] H. Barnum, J. Barrett, M. Leifer, and A. Wilce, *Generalized no-broadcasting theorem*, Phys. Rev. Lett. **99**, 240501 (2007), arXiv:0707.0620.
- [13] B. Coecke, D. Pavlović, and J. Vicary, *A new description of orthogonal bases*, Math. Structures Comput. Sci. **23**, 555 (2013), arXiv:0810.0812.
- [14] J. Vicary, *Categorical formulation of finite-dimensional quantum algebras*, Comm. Math. Phys. **304**, 765 (2011), arXiv:0805.0432.
- [15] H. Barnum and J. Hilgert, *Strongly symmetric spectral convex bodies are Jordan algebra state spaces*, arXiv:1904.03753.
- [16] R. Dijkgraaf, *A geometrical approach to two-dimensional conformal field theory*, Ph.D. thesis, Utrecht University (1989).
- [17] L. Abrams, *Two-dimensional topological quantum field theories and Frobenius algebras*, J. Knot Theory Ramifications **5**, 569 (1996).
- [18] J. Kock, *Frobenius Algebras and 2D Topological Quantum Field Theories*, London Math. Soc. Student Texts **59** (Cambridge University Press, 2004).